Modular Inverses and CRT

COMP2711 Discrete Mathematical Tools

Mingxun Zhou (mingxunz@ust.hk)

From GCDs to simultaneous equations

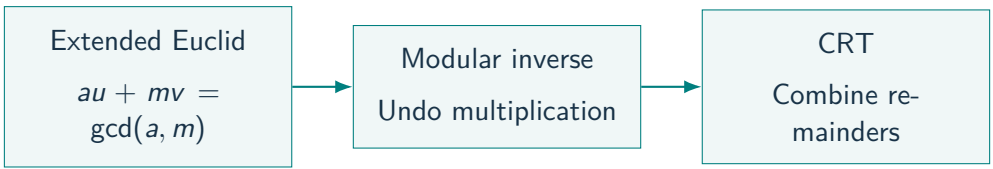
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All numbers are integers; throughout this lecture, moduli are greater than 1.

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Division is not always valid:

$$2 \times 1 \equiv 2 \times 4 \pmod{6}, \qquad 1 \not\equiv 4 \pmod{6}.$$

Review: GCD and Euclid

$\gcd(a, b)$ is the greatest positive common divisor of a, b (not both zero).
If $\gcd(a, b) = 1$, we say a, b are **coprime**.

Euclidean Algorithm for GCD:

$$\gcd(a, b) = \gcd(a \bmod b, b) \qquad (a \geq 0,\ \ b > 0).$$

Modular inverses and equations

Can we solve this equation?

In elementary algebra, over the real numbers:

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What should "divide by 17" mean modulo 43?

Modular inverse

MUST UNDERSTAND Modular inverse

An **inverse** of a modulo m is an integer u such that

$$au \equiv 1 \pmod{m}.$$

Notation: $u \equiv a^{-1} \pmod{m}$.

This plays the role of $1/a$ in modular arithmetic, when it exists.

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Example: $3 \times 5 = 15 \equiv 1 \pmod{7}$, so $5 \equiv 3^{-1} \pmod{7}$.

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Given a and m, we want an integer u such that $au \equiv 1 \pmod{m}$.

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Run Euclid first; then work backward to recover the coefficients.

Extended Euclid: 1. Run Euclid

a	b
43	17

Recall

$$\gcd(a, b) = \gcd(b, a \bmod b)$$

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Recall

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Extended Euclid: 2. The key backward step

Suppose we know a Bézout solution pair—integers s, t—for the smaller pair $(b, a \bmod b)$, such that

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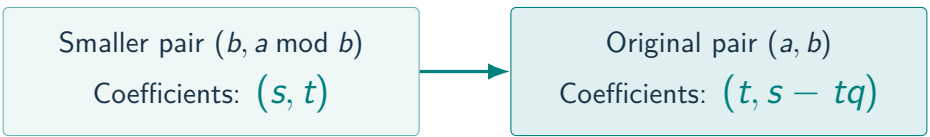
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$$ta + (s - tq)b = 1.$$



Extended Euclid: 3. Work backward through the table

In each row, find s, t such that $sa + tb = 1$.

a	b	$q = \lfloor a/b \rfloor$	s	t
43	17	2		
17	9	1		
9	8	1		
8	1	8		
1	0	—		

Original pair (a, b)

Coefficients: $(t, s - tq)$



Smaller pair $(b, a \bmod b)$

Coefficients: (s, t)

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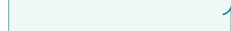
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1	**0**	—	1	0

Start at $(1, 0)$: choose $(s, t) = (1, 0)$.

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$$1 \times 1 + 0 \times 0 = 1.$$

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Using $q = 8$: $(s, t) = (0, 1 - 0 \times 8) = (0, 1)$.

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$$1 \times 9 + (-1) \times 8 = 1.$$

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$$(-1) \times 17 + 2 \times 9 = 1.$$

Extended Euclid: 3. Work backward through the table

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43	**17**	2	**2**	-5
17	9	1	-1	**2**
9	8	1	1	-1
8	1	8	0	1
1	0	—	1	0

Using $q = 2$: $(s, t) = (2, -1 - 2 \times 2) = (2, -5)$.

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Using $q = 2$: $(s, t) = (2, -1 - 2 \times 2) = (2, -5)$.

$$2 \times 43 + (-5) \times 17 = 1.$$

Extended Euclid: 4. Read the answer

The top row gives

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Return to our original goal: find u such that $17u \equiv 1 \pmod{43}$. We now know

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$$(-5) \times 17 = 1 - 2 \times 43.$$

$$(-5) \times 17 \equiv 1 \pmod{43}.$$

Thus our inverse is

$$\boxed{-5 \equiv 38 \equiv 17^{-1} \pmod{43}.}$$

Bézout's identity

Theorem (Bézout)

For integers a, m not both zero, there exist integers u, k such that

$$au + km = \gcd(a, m).$$

*The **Extended Euclidean Algorithm** finds such a pair (u, k).*

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The proof is nearly identical to the extended Euclidean argument: work backward from $(g, 0)$, where $g = \gcd(a, m)$.

The full proof is in the course note.

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*The **Extended Euclidean Algorithm** finds such a pair (u, k).*

Corollary (Modular inverses and coprimality)

For an integer a and $m > 1$,

$$\begin{aligned} & a \text{ has an inverse modulo } m \\ \iff\ & \text{there exist integers } u, k \text{ such that } au + km = 1 \\ \iff\ & \gcd(a, m) = 1. \end{aligned}$$

Solving the opening equation

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Multiply both sides by 38:

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$$x \equiv 190 \equiv 18 \pmod{43}.$$

$$\boxed{x = 18 + 43k, \qquad k \in \mathbb{Z}.}$$

Check: $17 \times 18 = 306 = 7 \times 43 + 5$.

Cancellation with an inverse

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Why the inverse is unique modulo m

If $au \equiv 1$ and $av \equiv 1 \pmod{m}$, then

$$au \equiv av \pmod{m} \implies u \equiv v \pmod{m}.$$

Non-coprime equations: The idea

If $g = \gcd(a, m) > 1$, try reducing to the coprime case.

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For example, divide the coefficient, right-hand side, and modulus by 2:

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The solutions are $x \equiv 6, 13 \pmod{14}$; the optional section gives the full procedure.

Combining congruences

Sunzi's remainder problem

《孫子算經》卷下，第26題

Traditional Chinese	Mathematical formulation
今有物不知其數，	Let x be the number of objects.
三三數之賸二，	$x \equiv 2 \pmod 3$.
五五數之賸三，	$x \equiv 3 \pmod 5$.
七七數之賸二，	$x \equiv 2 \pmod 7$.
問物幾何？	Find x.

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答曰：「二十三。」 Answer: 23.

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Chinese remainder theorem

MUST UNDERSTAND Chinese remainder theorem

Let $m_1, \ldots, m_n > 1$ be pairwise coprime: $\gcd(m_i, m_j) = 1$ for $i \neq j$. For any integers $r_1, \ldots, r_n$, the system

$$\begin{aligned} x &\equiv r_1 \pmod{m_1}, \\ &\vdots \\ x &\equiv r_n \pmod{m_n} \end{aligned}$$

has exactly one solution modulo $M = m_1 \cdots m_n$.

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We start with the case of two congruences.

Chinese remainder theorem: Two congruences

Theorem

Let $m_1, m_2 > 1$ *with* $\gcd(m_1, m_2) = 1$. *For any integers* r_1, r_2, *the system*

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has exactly one solution modulo $m_1 m_2$.

For more congruences: Combine the 1st and 2nd, then combine with the 3rd, the 4th, ...

CRT construction: 1. Use Bézout's identity

Bézout gives integers u_1, u_2 with

$$u_1 m_1 + u_2 m_2 = 1.$$

$$\gcd(m_1, m_2) = 1$$
$$x \equiv r_1 \pmod{m_1}$$
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Raw term	Modulo m_1	Modulo m_2
$r_2 u_1 m_1$	0	r_2

CRT construction: 2. Match the remainders

Similarly, multiply Bézout's identity by r_1:

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Raw term	Modulo m_1	Modulo m_2
$r_2 u_1 m_1$	0	r_2
$r_1 u_2 m_2$	r_1	0
$x \equiv r_2 u_1 m_1 + r_1 u_2 m_2 \pmod{m_1 m_2}$	r_1	r_2

CRT construction: 3. Work through the example

$$x \equiv 2 \pmod 3,$$
$$x \equiv 3 \pmod 5.$$

$\gcd(m_1, m_2) = 1$
$x \equiv r_1 \pmod{m_1}$
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CRT construction: 3. Work through the example

$$x \equiv 2 \pmod 3,$$
$$x \equiv 3 \pmod 5.$$

Bézout: $2 \times 3 + (-1) \times 5 = 1$, so $u_1 = 2$, $u_2 = -1$.

$\gcd(m_1, m_2) = 1$
$x \equiv r_1 \pmod{m_1}$
$x \equiv r_2 \pmod{m_2}$

Raw term	Modulo 3	Modulo 5
$r_2 u_1 m_1 = 3 \times 2 \times 3 = 18$	0	3
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Bézout: $2 \times 3 + (-1) \times 5 = 1$, so $u_1 = 2$, $u_2 = -1$.

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All integer solutions: $x = 8 + 15k$, $k \in \mathbb{Z}$.

CRT: Why the solution is unique

Assume x and y are any two solutions:

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If a and b are coprime, i.e., $\gcd(a, b) = 1$, and both divide x, then $ab \mid x$.

Proof: exercise; see the course note.

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Thus $m_1 m_2 \mid (x - y)$, so $x - y \equiv 0 \pmod{m_1 m_2}$.

Optional: The general CRT solution

For pairwise-coprime $m_1, \ldots, m_n > 1$, solve $x \equiv r_i \pmod{m_i}$.

$$M = \prod_{i=1}^{n} m_i, \qquad M_i = \frac{M}{m_i}.$$

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Modulo m_j, only the jth term survives, giving remainder r_j.

Class 1: Summary

1. Extended Euclid finds u, v with $au + mv = \gcd(a, m)$.
2. If $\gcd(a, m) = 1$, then $u \equiv a^{-1} \pmod{m}$.
3. CRT combines prescribed remainders for pairwise-coprime moduli into one solution modulo their product.

Review exercises: Inverses and congruences

1. Modular inverse. Use extended Euclid to find $12^{-1} \pmod{35}$. Then solve

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1. Modular inverse. Use extended Euclid to find $12^{-1} \pmod{35}$. Then solve

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2. Chinese remainder theorem. Find all integers x satisfying

$$\begin{aligned} x &\equiv 2 \pmod{3}, \\ x &\equiv 3 \pmod{5}, \\ x &\equiv 2 \pmod{7}. \end{aligned}$$

Optional: Non-coprime equations

When there is no inverse

$\gcd(6, 14) = 2$, so 6 has no inverse modulo 14.

Can either equation still have solutions?

$$6x \equiv 8 \pmod{14} \qquad 6x \equiv 7 \pmod{14}$$

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The second would require $6x - 14k = 7$ for some integer k.

$$\underbrace{6x - 14k}_{\text{even}} \neq \underbrace{7}_{\text{odd}}.$$

No solution to $6x \equiv 7 \pmod{14}$**.**

Non-coprime case: Reduce the modulus

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4. Set $m' = m/g$. If $m' > 1$, find

$$x_0 \equiv (a/g)^{-1}(b/g) \pmod{m'}, \qquad 0 \le x_0 < m'.$$

If $m' = 1$, take $x_0 = 0$.

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Let $g = \gcd(a, m)$ with $g \mid b$, and set $m' = m/g$.

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If $m' = 1$, take $x_0 = 0$: every integer x is a solution.

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0 1 2 3 4 5 (6) 7 8 9 10 11 12 (13)

$$\boxed{x \equiv 6, 13 \pmod{14}.}$$

Linear congruences: The general rule

Theorem

Let $g = \gcd(a, m)$. For $ax \equiv b \pmod{m}$:

- *solutions exist exactly when $g \mid b$;*
- *if solvable, there are exactly g solutions among $0, \ldots, m-1$.*

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Equation	GCD	Number of solutions
$17x \equiv 5 \pmod{43}$	1	1
$6x \equiv 8 \pmod{14}$	2	2
$6x \equiv 7 \pmod{14}$	2	0

Practice: Zero or several solutions?

Solve each congruence:

$$15x \equiv 9 \pmod{24}, \qquad 15x \equiv 10 \pmod{24}.$$

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$$15x \equiv 9 \pmod{24}, \qquad 15x \equiv 10 \pmod{24}.$$

1. Predict the number of solutions using the GCD.
2. For each solvable equation, list every solution among $0, \ldots, 23$.

Full solutions in the note.